

Student Management System

Overview

Student Management System is a REST API developed using Spring Boot, Spring Data JPA, Hibernate, and MySQL. The application enables administrators to manage students, courses, addresses, and enrollments efficiently.

Technologies Used

- Java 17
 - Spring Boot 3
 - Spring Data JPA
 - Hibernate
 - MySQL
 - Spring Security
 - Swagger OpenAPI
 - Lombok
 - Maven
 - JUnit 5
-

Features

Admin Operations

Student Admission

- Add new students
- Store unique student code
- Store personal information
- Add multiple addresses for a student

Course Management

- Create courses
- View all courses
- Get course details by ID

Course Assignment

- Assign courses to students
- Remove assigned courses from students

Search Functionality

- Search students by name
- Get students assigned to a specific course

- Search courses by topic
-

Student Operations

Profile Management

Students can update:

- Email
- Mobile Number
- Parent Name
- Address Information

Student Validation

Students can validate their identity using:

- Student Code
- Date of Birth

Course Enrollment

- View assigned courses
 - Leave assigned courses
-

Entity Relationships

Student ↔ Address

One-to-Many Relationship

- One student can have multiple addresses.
- One address belongs to one student.

Student ↔ Course

Many-to-Many Relationship

- One student can enroll in multiple courses.
 - One course can be assigned to multiple students.
-

API Endpoints

Student APIs

Method	Endpoint
POST	/students
GET	/students
GET	/students/{id}
PUT	/students/{id}
DELETE	/students/{id}
GET	/students/search?name=value
POST	/students/validate

Course APIs

Method	Endpoint
POST	/courses
GET	/courses
GET	/courses/{id}
GET	/courses/topic?topic=value

Address APIs

Method	Endpoint
POST	/addresses
GET	/addresses

Enrollment APIs

Method	Endpoint
POST	/students/{studentId}/courses/{courseId}
DELETE	/students/{studentId}/courses/{courseId}
GET	/students/course/{courseId}
GET	/students/{studentId}/courses

Database Configuration

Update `application.properties` :

```
spring.datasource.url=jdbc:mysql://localhost:3306/studentdb
spring.datasource.username=root
spring.datasource.password=your_password

spring.jpa.hibernate.ddl-auto=update
spring.jpa.show-sql=true
```

Running the Application

Clone the repository:

```
git clone <repository-url>
```

Navigate to project directory:

```
cd student-management-system
```

Run the application:

```
mvn spring-boot:run
```

Application will start on:

```
http://localhost:8080
```

Swagger Documentation

Swagger UI:

```
http://localhost:8080/swagger-ui/index.html
```

Testing

Run unit tests:

```
mvn test
```

Project Structure

```
src
├── main
│   ├── java
│   │   ├── controller
│   │   ├── service
│   │   ├── repository
│   │   ├── entity
│   │   ├── dto
│   │   └── config
│   └── resources
│       └── application.properties
└── test
    └── java
```

Author

Backend Assignment Submission

Developed using Spring Boot, JPA, Hibernate, MySQL, Swagger, and Spring Security.